Applications of the Dot Product

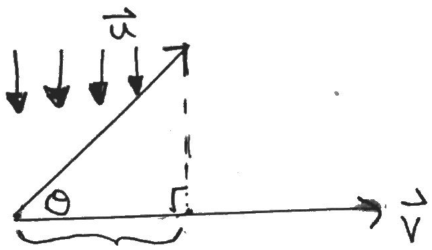
Recall: $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cos(\theta)$ (geometric)

$= u_1 v_1 + u_2 v_2$ (component)

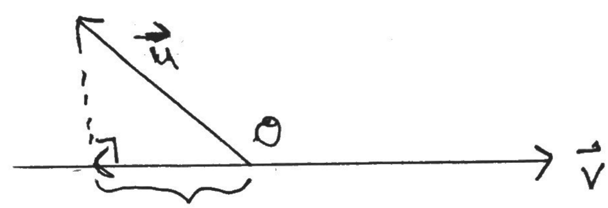
Projections

A projection is formed by dropping a $\perp$ from each point of an object onto a line or plane.

An example of this is the shadow of an object.



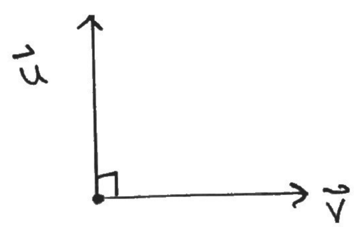
$0 < \theta < \frac{\pi}{2}$; proj ($\vec{u}$ onto $\vec{v}$)



$\frac{\pi}{2} < \theta < \pi$; proj ($\vec{u}$ onto $\vec{v}$)

NOTES: ① If $\frac{\pi}{2} < \theta < \pi$ then the direction of $\text{proj}_{\vec{v}} \vec{u}$ is opposite to the direction of $\vec{v}$

② If $\vec{u} \perp \vec{v}$, then $\vec{u}$ casts no shadow



$\Rightarrow \text{proj}_{\vec{v}} \vec{u} = \vec{0}$

The vector obtained by projecting $\vec{u}$ $\perp$ onto the line through $\vec{v}$ is called the vector projection of $\vec{u}$ onto $\vec{v}$.

* $\text{proj}_{\vec{v}}\vec{u}$ is a vector.
* $|\text{proj}_{\vec{v}}\vec{u}|$ is the magnitude of this vector.

* $$\text{proj}_{\vec{v}}\vec{u} = |\vec{u}|\cos(\theta)\cdot\hat{v}$$
$$= |\vec{u}|\cos(\theta)\cdot\frac{\vec{v}}{|\vec{v}|}$$
$$= |\vec{u}||\vec{v}|\cos(\theta)\cdot\frac{\vec{v}}{|\vec{v}|^2}$$

* $$\text{proj}_{\vec{v}}\vec{u} = (\vec{u}\bullet\vec{v})\cdot\frac{\vec{v}}{|\vec{v}|^2}$$

$$\text{proj}_{\vec{v}}\vec{u} = \frac{(\vec{u}\bullet\vec{v})\,\vec{v}}{\vec{v}\bullet\vec{v}}$$

* $$|\text{proj}_{\vec{v}}\vec{u}| = |\vec{u}|\cos(\theta)$$

Ex 1: Given $\vec{u} = [3, 6]$, $\vec{v} = [-4, 3]$, find $\text{proj}_{\vec{v}} \vec{u}$ and $\text{proj}_{\vec{u}} \vec{v}$.

a) $\text{proj}_{\vec{v}} \vec{u} = \frac{(\vec{u} \bullet \vec{v})}{|\vec{v}|^2} \cdot \vec{v}$

NOTE:

$\vec{u} \bullet \vec{v} = -12 + 18$

$= 6$

$|\vec{v}| = \sqrt{(-4)^2 + (3)^2}$

$|\vec{v}| = 5$

$= \frac{6}{25} \cdot \vec{v}$

$= \frac{6}{25} [-4, 3]$

$= \left[\frac{-24}{25}, \frac{18}{25} \right]$

b) $\text{proj}_{\vec{u}} \vec{v} = \frac{(\vec{v} \bullet \vec{u})}{|\vec{u}|^2} \cdot \vec{u}$

NOTE:

$\vec{v} \bullet \vec{u} = 6$

$|\vec{u}| = \sqrt{(3)^2 + (6)^2}$

$|\vec{u}| = \sqrt{45}$

$= \frac{6}{45} \vec{u}$

$= \frac{2}{15} \cdot \vec{u}$

$= \frac{2}{15} [3, 6]$

$= \left[\frac{2}{5}, \frac{4}{5} \right] = \frac{2}{5} [1, 2]$

Work

Work is done whenever a force acting on an object causes a displacement of the object from one position to another.

* The "work" done by a force is defined as the dot product. Units are in Joules (J)

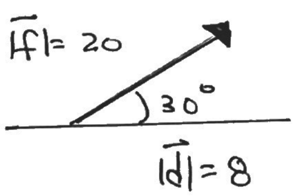
i.e.: $W = \vec{f} \bullet \vec{d}$ ← displacement caused by the force.

work — force acting on an object

$$W = |\vec{f}| \cdot |\vec{d}| \cos(\theta)$$

θ = angle between the force and displacement vectors.

Ex 1: A crate is hauled 8 m under a constant force of 20 N that is applied at an angle of 30°. Determine the work done.



$$W = |\vec{f}||\vec{d}|\cos(\theta)$$

$$W = (20)(8)\cos(30^\circ)$$

$$W \doteq 138.6 \text{ J}.$$

Ex 2: A box is pulled 5 m along a smooth level floor by a 40 N force that is applied at 28° to the floor. The box is then pulled 2 m up a ramp inclined at 22° to the horizontal, using the same force. The box is finally pulled a further 3 m along a level platform using the same force. Determine the work done in moving the box.

Solution:

$$W = \vec{F} \bullet \vec{d}$$

$$W = |\vec{F}||\vec{d}| \cos(\theta)$$

$$W = (40)(5)\cos(28^\circ) + (40)(2)\cos(6^\circ) + (40)(3)\cos(28^\circ)$$

$$W \doteq 176.59 + 79.56 + 105.95$$

$$W \doteq 362.1 \text{ J}$$

Ex 3: A force of 50 N is acting in the direction of $\vec{v} = [3, 2]$.

a) Determine a unit vector in the direction of $\vec{v}$.

Solution:

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|} = \frac{[3,2]}{\sqrt{13}} = \left[\frac{3}{\sqrt{13}}, \frac{2}{\sqrt{13}}\right]$$

b) Determine the force vector in the direction of $\vec{v}$.

Solution:

$$\vec{F} = 50 \cdot \hat{v}$$

$$\vec{F} = \left[\frac{150}{\sqrt{13}}, \frac{100}{\sqrt{13}}\right] = \frac{50}{\sqrt{13}}[3,2]$$

c) The force is exerted on an object moving it from the point $(3,0)$ to $(14,0)$ with distance in metres. Determine the work done.

Solution: $W = \vec{F} \bullet \vec{d}$

$$W = \left[\frac{150}{\sqrt{13}}, \frac{100}{\sqrt{13}}\right] \bullet [11, 0]$$

$$W = \left(\frac{150}{\sqrt{13}}\right)(11) + \cancelto{0}{\left(\frac{100}{\sqrt{13}}\right)(0)}$$

$$W = \frac{1650}{\sqrt{13}}$$

$$W \doteq 457.63 \text{ J}$$